

Introduction to Microeconomics

Lecture 2—Comparative Advantage

Prof. Alejandro Melo-Ponce



Years
Anniversary

ECON101
Spring 2026

Comparative Advantage: The Basis of Exchange

Why do people exchange goods and services in the first place? Why not just produce our own food, cars, clothing, shelter, and the like? The answer is that we can all have more of every good and service if we specialize in the activities at which we are **relatively** most efficient.

Comparative Advantage: The Basis of Exchange

Why do people exchange goods and services in the first place? Why not just produce our own food, cars, clothing, shelter, and the like? The answer is that we can all have more of every good and service if we specialize in the activities at which we are **relatively** most efficient.

How much would it cost to produce a chicken sandwich from scratch?

According to this YouTuber^a, it took him 6 months of his life and \$1500 to make a sandwich from scratch! [▶ Link](#)

⚠ Viewer discretion advised! The video shows the murder of two chicken. ☹

^aYou should always be careful with statements said by social media creators. Some of them are knowledgeable and produce good content, but many of them are very ignorant people (that is why they became social media creators in the first place) whose only incentives are to attract the largest number of views by creating click-bait trash content. Use your critical thinking skills before believing what you hear and see in social media!

Akmaral and Sabina

	Painting	Roofing
Akmaral	300 hours	400 hours
Sabina	200 hours	100 hours

- Sabina may say that her doing both jobs is unfair. A more telling objection is that it is **inefficient**.

Akmaral and Sabina

	Painting	Roofing
Akmaral	300 hours	400 hours
Sabina	200 hours	100 hours

- Sabina may say that her doing both jobs is unfair. A more telling objection is that it is **inefficient**.
- The reason is that Akmaral has a **comparative advantage** over Sabina in painting, which means that she is **relatively** more efficient at painting than Sabina.

Akmaral and Sabina

	Painting	Roofing
Akmaral	300 hours	400 hours
Sabina	200 hours	100 hours

- Sabina may say that her doing both jobs is unfair. A more telling objection is that it is **inefficient**.
- The reason is that Akmaral has a **comparative advantage** over Sabina in painting, which means that she is **relatively** more efficient at painting than Sabina.
- Key word: **relatively**.

Akmaral and Sabina

	Painting	Roofing
Akmaral	300 hours	400 hours
Sabina	200 hours	100 hours

- Sabina may say that her doing both jobs is unfair. A more telling objection is that it is **inefficient**.
- The reason is that Akmaral has a **comparative advantage** over Sabina in painting, which means that she is **relatively** more efficient at painting than Sabina.
- Key word: **relatively**.
- This means that Akmaral has a **lower opportunity cost** of painting than Sabina does.

Akmaral and Sabina

	Painting	Roofing
Akmaral	300 hours	400 hours
Sabina	200 hours	100 hours

- Sabina may say that her doing both jobs is unfair. A more telling objection is that it is **inefficient**.
- The reason is that Akmaral has a **comparative advantage** over Sabina in painting, which means that she is **relatively** more efficient at painting than Sabina.
- Key word: **relatively**.
- This means that Akmaral has a **lower opportunity cost** of painting than Sabina does.
- Akmaral: the opportunity cost of painting a house is 0.75 roofing jobs.

Akmaral and Sabina

	Painting	Roofing
Akmaral	300 hours	400 hours
Sabina	200 hours	100 hours

- Sabina may say that her doing both jobs is unfair. A more telling objection is that it is **inefficient**.
- The reason is that Akmaral has a **comparative advantage** over Sabina in painting, which means that she is **relatively** more efficient at painting than Sabina.
- Key word: **relatively**.
- This means that Akmaral has a **lower opportunity cost** of painting than Sabina does.
- Akmaral: the opportunity cost of painting a house is 0.75 roofing jobs.
- Sabina: the opportunity cost of painting a house is 2 roofing jobs.

Akmaral and Sabina

	Painting	Roofing
Akmaral	300 hours	400 hours
Sabina	200 hours	100 hours

- Sabina may say that her doing both jobs is unfair. A more telling objection is that it is **inefficient**.
- The reason is that Akmaral has a **comparative advantage** over Sabina in painting, which means that she is **relatively** more efficient at painting than Sabina.
- Key word: **relatively**.
- This means that Akmaral has a **lower opportunity cost** of painting than Sabina does.
- Akmaral: the opportunity cost of painting a house is 0.75 roofing jobs.
- Sabina: the opportunity cost of painting a house is 2 roofing jobs.
- Akmaral has a comparative advantage in painting, and it makes sense for her to do both painting jobs, leaving the two roofing jobs to Sabina.

	Painting	Roofing
Akmaral	300 hours	400 hours
Sabina	200 hours	100 hours

- Sabina may say that her doing both jobs is unfair. A more telling objection is that it is **inefficient**.
- The reason is that Akmaral has a **comparative advantage** over Sabina in painting, which means that she is **relatively** more efficient at painting than Sabina.
- Key word: **relatively**.
- This means that Akmaral has a **lower opportunity cost** of painting than Sabina does.
- Akmaral: the opportunity cost of painting a house is 0.75 roofing jobs.
- Sabina: the opportunity cost of painting a house is 2 roofing jobs.
- Akmaral has a comparative advantage in painting, and it makes sense for her to do both painting jobs, leaving the two roofing jobs to Sabina.
- If each performed both the painting and roofing tasks, the total time spent would be 700 hours for Akmaral and 300 hours for Sabina. By **specializing**, the total work time falls to 600 hours for Akmaral and 200 hours for Sabina, saving 100 hours each.

Rule

To compute the opportunity cost of good A in units of good B :

$OC(A \text{ in } B) = \text{Cost of 1 unit of } A \text{ in terms of units of } B \text{ given up}$

$$\begin{aligned} &= 1 \text{ unit of } A \left(\frac{\text{Cost of } x \text{ units in resource units}}{x \text{ units of } A} \right) \left(\frac{y \text{ units of } B}{\text{Cost of } y \text{ units in resource units}} \right) \\ &= \frac{y}{x}. \end{aligned}$$

Always write units explicitly and cancel them. If units cancel correctly, the calculation is correct.

Akmaral and Sabina: Data (hours per job)

	Painting	Roofing
Akmaral	300	400
Sabina	200	100

Opportunity cost of one painting job (in roofing jobs)

Akmaral:

$$OC_A(P \text{ in } R) = (1P) \left(\frac{300 \text{ hours}}{1P} \right) \left(\frac{1R}{400 \text{ hours}} \right) = 0.75 R.$$

Sabina:

$$OC_S(P \text{ in } R) = (1P) \left(\frac{200 \text{ hours}}{1P} \right) \left(\frac{1R}{100 \text{ hours}} \right) = 2 R.$$

Conclusion

Akmaral has a **lower opportunity cost** of painting than Sabina and therefore a **comparative advantage** in painting. Sabina has a comparative advantage in roofing.

The principle of Comparative Advantage

Everyone does best when each person (or country) concentrates on the activities in which he or she is relatively most efficient.

⚠ Caution!

Pay close attention to the form in which the productivity information is provided. Your goal in each case is to find each person's opportunity cost of producing the good in question.

Sezim and Nazima

Sezim can edit a short video in 10 minutes and write a social media post in 4 minutes. Nazima can edit a short video in 6 minutes and write a social media post in 3 minutes. Which statement below is true?

- ① Sezim has no comparative advantage.
- ② Nazima should do both tasks because she has an absolute advantage in both.
- ③ Sezim has a comparative advantage in writing posts and Nazima has a comparative advantage in video editing.
- ④ Sezim has a comparative advantage in video editing and Nazima has a comparative advantage in writing posts.
- ⑤ None of the above statements is true.

Sezim and Nazima (Minutes per Task)

Sezim: 10 min per video, 4 min per post.

Nazima: 6 min per video, 3 min per post.

Opportunity cost of one video (in posts)

Sezim:

$$OC_S(V \text{ in } P) = (1V) \left(\frac{10 \text{ min}}{1V} \right) \left(\frac{1P}{4 \text{ min}} \right) = 2.5P.$$

Nazima:

$$OC_N(V \text{ in } P) = (1V) \left(\frac{6 \text{ min}}{1V} \right) \left(\frac{1P}{3 \text{ min}} \right) = 2P.$$

Conclusion

Nazima has a **lower opportunity cost** of producing videos than Sezim and therefore a **comparative advantage** in video editing.

Sezim and Nazima

Sezim can edit 10 short videos per hour or write 4 social media posts per hour. Nazima can edit 6 short videos per hour or write 3 social media posts per hour. Which statement below is true?

- ① Sezim has no comparative advantage.
- ② Nazima should do both tasks because she has an absolute advantage in both.
- ③ Sezim has a comparative advantage in writing posts and Nazima has a comparative advantage in video editing.
- ④ Sezim has a comparative advantage in video editing and Nazima has a comparative advantage in writing posts.
- ⑤ None of the above statements is true.

Sezim and Nazima (tasks per hour)

Sezim: 10 videos/hour, 4 posts/hour.

Nazima: 6 videos/hour, 3 posts/hour.

Opportunity cost of one video (in posts)

Sezim:

$$OC_S(V \text{ in } P) = (1V) \left(\frac{1 \text{ hour}}{10V} \right) \left(\frac{4P}{1 \text{ hour}} \right) = 0.4P.$$

Nazima:

$$OC_N(V \text{ in } P) = (1V) \left(\frac{1 \text{ hour}}{6V} \right) \left(\frac{3P}{1 \text{ hour}} \right) = 0.5P.$$

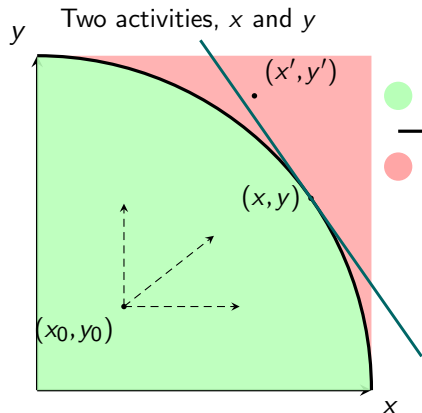
Conclusion

Sezim has a **lower opportunity cost** of producing videos than Nazima and therefore a **comparative advantage** in video editing.

The production possibilities frontier (PPF)

Production Possibilities Frontier

It is the boundary between those combinations of goods and services that can be produced, given current resources, and those that cannot.



- feasible but inefficient
- feasible & efficient
- infeasible

$|\text{slope}| = \text{C.O. of 1 more unit of } x \text{ in terms of units of } y \text{ lost.}$

$\left| \frac{1}{\text{slope}} \right| = \text{C.O. of 1 more } y \text{ in terms of units of } x \text{ lost.}$

Ali on a Remote Farm

Ali has ended up alone on a remote Stardew Valley–style farm, completely isolated from the rest of the world. He must divide his time between building shelter and producing food. In one week, Ali can produce either 6 m^2 /week of shelter or 12 kg/week of food. Describe Ali's PPF.

Opportunity cost of food

$$OC_A(f \text{ in } s) = (1f) \left(\frac{1 \text{ week}}{12f} \right) \left(\frac{6s}{1 \text{ week}} \right) = 0.5s.$$

PPF

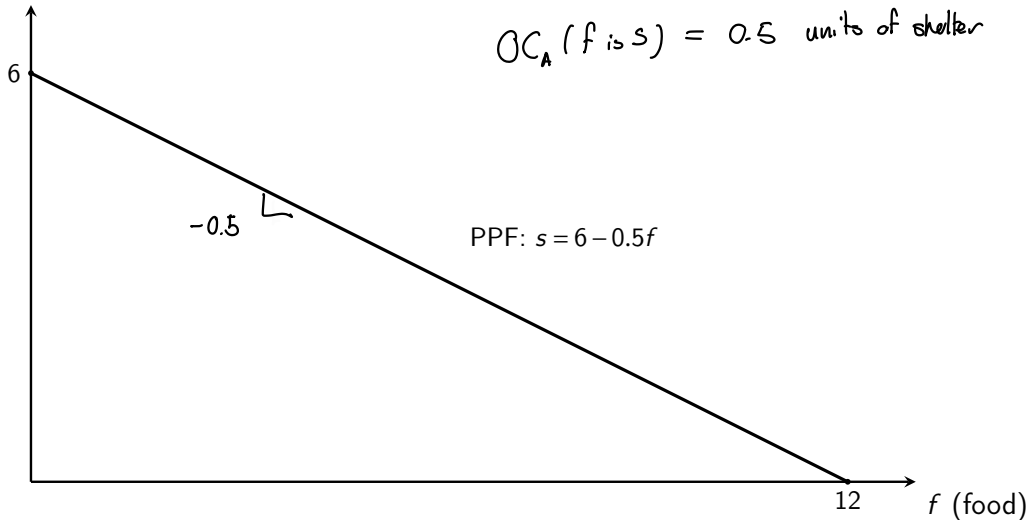
Ali's PPF is linear and given by

$$s = 6 - 0.5f,$$

with intercepts $(f, s) = (12, 0)$ and $(0, 6)$.

Ali's Production Possibility Frontier

s (shelter)



Medina on Her Own Plot

Medina is also isolated on a nearby plot of land. If she devotes all her time to shelter, she can build 4 m^2 /week. If she devotes all her time to food production, she can produce 4 kg/week of food. Describe Medina's PPF.

Opportunity cost of food

$$OC_M(f \text{ in } s) = (1f) \left(\frac{1 \text{ week}}{4f} \right) \left(\frac{4s}{1 \text{ week}} \right) = 1s.$$

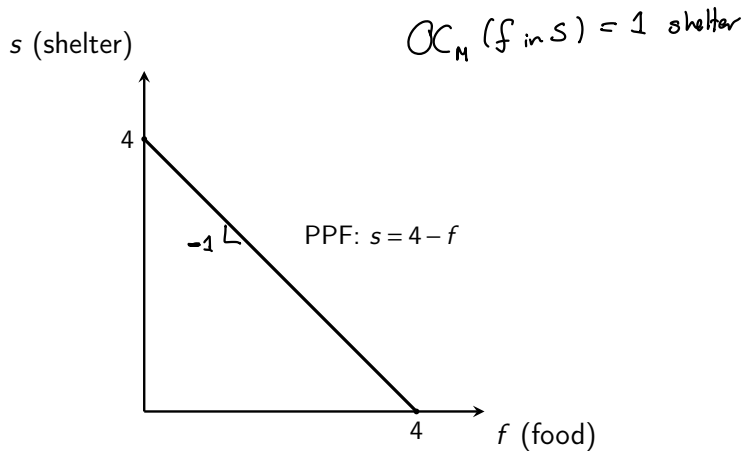
PPF

Medina's PPF is linear and given by

$$s = 4 - f,$$

with intercepts $(f, s) = (4, 0)$ and $(0, 4)$.

Medina's Production Possibility Frontier



PPF in a Two-Person Economy

Nuray, who naturally takes charge, convinces Ali and Medina to coordinate their efforts. Even though they remain isolated from the rest of the world, they now plan production jointly. In the examples above, Ali has a comparative advantage in producing food, while Medina has a comparative advantage in producing shelter.

Ali and Medina under Nuray's Organization

Ali can produce 6 m^2 /week of shelter or 12 kg/week of food. Medina can produce 4 m^2 /week of shelter or 4 kg/week of food.

Describe the economy's PPF.

PPF in a Two-person farm: Solution

Individual technologies

In one week:

- Ali can produce either 12 f or 6 s .
- Medina can produce either 4 f or 4 s .

Comparative advantage

Ali has a lower opportunity cost of producing food; Medina has a lower opportunity cost of producing shelter.

Joint production possibilities

- If both specialize according to comparative advantage:

$$f = 12 \quad \text{and} \quad s = 4.$$

- If both specialize in shelter:

$$s = 6 + 4 = 10 \quad \text{and} \quad f = 0.$$

Ali: (12, 0)
Medina: (0, 4)

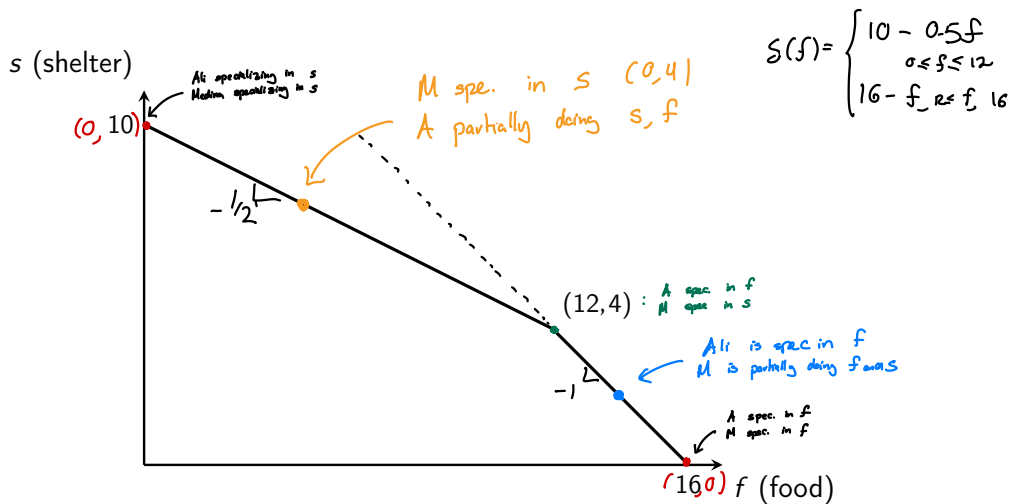
 (12, 4)

Joint PPF

The economy's PPF is piecewise linear with a kink at (12,4), connecting

$$(0, 10) \rightarrow (12, 4) \rightarrow (16, 0).$$

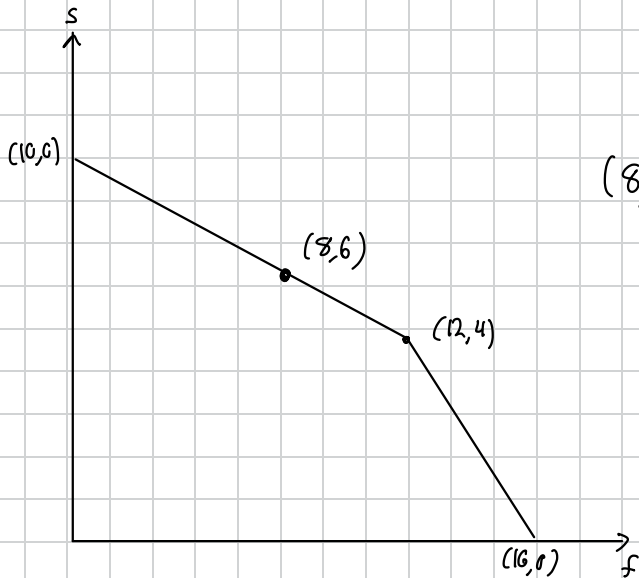
Joint Production Possibility Frontier



Production Assignment on the Farm

Under Nuray's direction, Ali and Medina decide to jointly consume $6 \text{ m}^2/\text{week}$ of shelter and 8 kg/week of food.

How should Nuray assign the production tasks between Ali and Medina?



$$s(f) = \begin{cases} 10 - 0.5f, & f \in [0, 12] \\ 16 - f, & f \in [12, 16] \end{cases}$$

$(8, 6)$: Medina is producing $(0, 4)$
Ali is producing $(8, 2)$

 $(8, 6)$

The Principle of Increasing Opportunity Cost

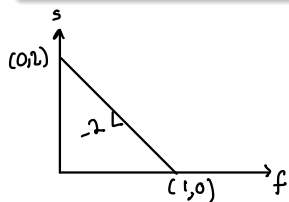
In expanding the production of any good, first employ those workers whose opportunity cost of producing that good is lowest, and only afterward turn to workers with higher opportunity costs.

Nuray Joins the Farm

Eventually, Nuray herself joins the production process. She can produce 2 m^2 /week of shelter or 1 kg/week of food.

Start by describing Nuray's individual PPF. Then describe the PPF of the three-person economy under her coordination.

If they jointly decide to produce 14 kg/week of food and 4 m^2 /week of shelter, how should Nuray allocate each person's work time?



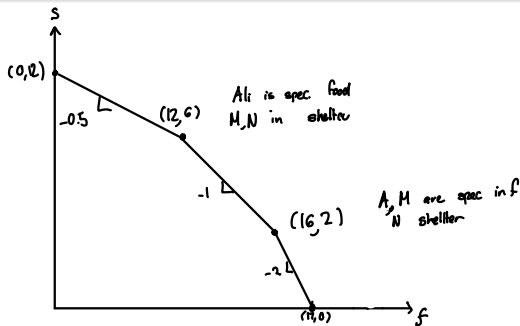
$$OC_H(f \text{ in } s) = 2s$$

$$OC_i(f \text{ in } s)$$

$$A: 0.5s$$

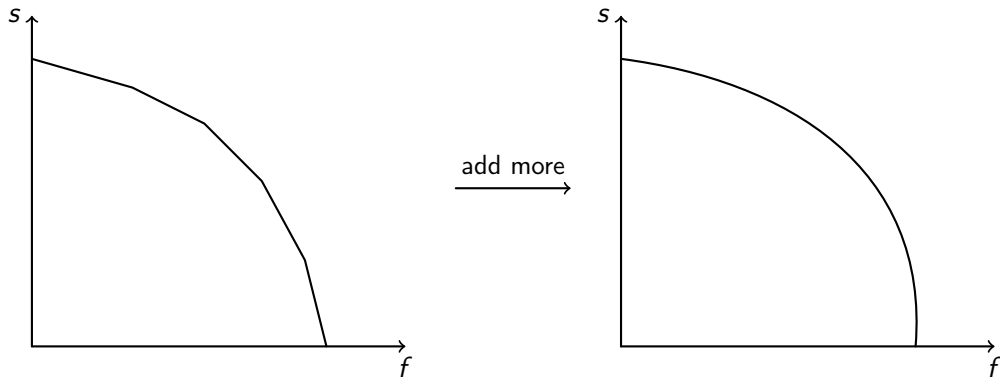
$$M: 1s$$

$$N: 2s$$



The production possibilities frontier with many workers

Intuition behind its shape: produce the initial units of clothing using the resources that are relatively most efficient at clothing production, and only then turn to those that are relatively less efficient at clothing production.



Another example of a bowed out (strictly concave) PPF

A Production Possibilities Frontier with Labor as the Only Input

Suppose that there are two goods, x and y , which require labor as an input of production. Let l_x and l_y denote the amounts of labor used in the production of x and y , respectively. Suppose that there are 100 hours of labor available.

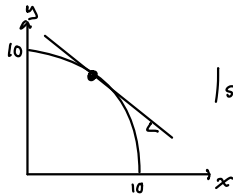
Assume also that the production processes of x and y are given by

$$x = \sqrt{l_x}, \quad y = \sqrt{l_y}.$$

Describe, mathematically and graphically, the production possibilities frontier (PPF).

Resource constraint: $100 \text{ hours} = l_x + l_y$

$$\left. \begin{array}{l} x = \sqrt{l_x} \quad x^2 = l_x \\ y = \sqrt{l_y} \quad y^2 = l_y \end{array} \right\} \Rightarrow \left\{ \begin{array}{l} 100 = x^2 + y^2 \\ x \geq 0, y \geq 0 \end{array} \right.$$
$$l_x, l_y \geq 0 \Rightarrow \begin{array}{l} x \geq 0 \\ y \geq 0 \end{array}$$



$$y = \sqrt{100 - x^2}$$
$$|\text{slope}| = \left| \frac{dy}{dx} \right| = OC(x \text{ in } y)$$

A second isolated farm

Erika, Bagdat, and Sultan

On a different isolated farm, three individuals decide to coordinate production.

Weekly production possibilities:

- Erika can produce 3 m^2 /week of shelter or 9 kg/week of food. $OC_E(f_{in s}) = 1/3 s$
- Bagdat can produce 8 m^2 /week of shelter or 4 kg/week of food. $OC_B(f_{in s}) = 2 s$
- Sultan can produce 6 m^2 /week of shelter or 6 kg/week of food. $OC_S(f_{in s}) = 1 s$

Sultan decides to organize the three of them and becomes the boss.

Describe the PPF of this three-person economy.

All shelter: (0, 17)

Erika food, B, S, shelter (9, 14)

Erika, Sultan food, B shelter (15, 6)

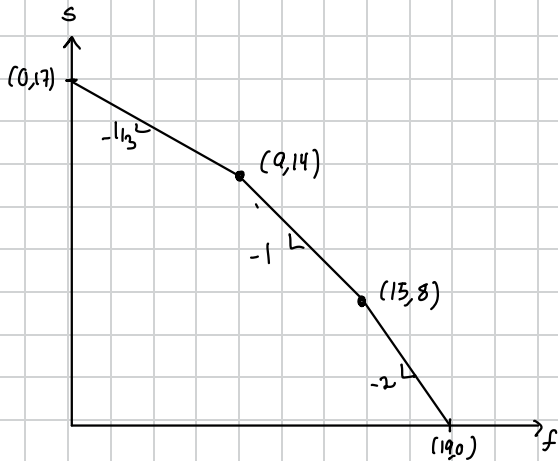
All food: (19, 0)

All shelter: $(0, 17)$

Evilke food, B, S, shelter $(9, 14)$

Evilke, Sultan food, B shelter $(15, 8)$

All food: $(19, 0)$



The two farms get organized—From isolation to interaction

Two farms discover each other

Until now, the two farms have been completely isolated.

Suddenly, they discover that the other farm exists.

Production technologies do not change. What changes is that they can now *trade* with each other.

Organizing into firms

As soon as trade becomes possible, something remarkable happens:

- Nuray organizes her farm and calls it **Nonchalant Empire™**.
- Sultan organizes his farm and calls it **Superior Farm™**.

Titles are assigned. Logos are imagined. PowerPoint decks are created.

Economically speaking, these are still just groups of workers with production technologies.

Nonchalant Empire: production possibilities

Weekly technology

Nonchalant Empire consists of three workers:

- Ali can produce 6 m^2 of shelter or 12 kg of food.
- Medina can produce 4 m^2 of shelter or 4 kg of food.
- Nuray can produce 2 m^2 of shelter or 1 kg of food.

Compute Nonchalant Empire's opportunity costs and identify its comparative advantage.

$$OC_{\text{Non-chalant}} (f \text{ in } s) = 0.5 \quad (\text{Ali is producing food first})$$

Superior Farm: production possibilities

Weekly technology

Superior Farm consists of three workers:

- Erika can produce 3 m^2 of shelter or 9 kg of food.
- Bagdat can produce 8 m^2 of shelter or 4 kg of food.
- Sultan can produce 6 m^2 of shelter or 6 kg of food.

Compute Superior Farm's opportunity costs and identify its comparative advantage.

$$OC_{SF} (f \text{ in } s) = \frac{1}{3} s \quad (\text{Erika produces food first})$$

∴ SF has comp. advantage in food

When is trade beneficial?

Terms of trade

Let p denote the **terms of trade** or more colloquially, the **price** of food in units of shelter:

$$p = \frac{m^2 \text{ of shelter}}{\text{kg of food}}.$$

Trade is mutually beneficial if p lies strictly between the opportunity costs of producing food in the two corporations:

$$OC^S(\text{food}) < p < OC^N(\text{food}).$$

Find such a value of p .

Predicting behavior

Suppose a price p satisfying the inequality has been chosen.

- Which corporation specializes relatively more in food?
- Which corporation specializes relatively more in shelter?
- Why does this follow from comparative advantage?

A concrete trade scenario

Choose a price $p =$ _____.

Suppose:

- Nonchalant Empire exports _____ m^2 of shelter.
- Superior Farm exports _____ kg of food.

Verify:

- Each firm can produce its export bundle.
- After trade, each firm can consume a bundle that was infeasible under autarky (no-trade).

Conclude: trade is beneficial.